

Disproof of conjecture 00000001006

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Verdict: FALSE

We disprove conjecture 00000001006 as stated. The conjecture asserts that an m -ovoid of the parabolic quadric $Q(4, q)$ can exist only when $m \mid (q + 1)$. At the smallest case $q = 2$, $m = 2$ the claim fails: $Q(4, 2)$ has an ovoid (a 1-ovoid), and the complement of that ovoid meets each of the 15 generating lines in exactly 2 points, so it is a 2-ovoid. But $2 \nmid 3 = q + 1$.

1 The conjecture

Quoted verbatim from `conjectures/00000001006.md`:

English. Definition: An m -ovoid (meeting every generating line in exactly m points).
Conjecture: m -ovoids of $Q(4, q)$ exist only when $m \mid (q + 1)$ (an existence criterion for m -ovoids).

中文。定义： m -ovoid(每条生成线恰交 m 点)。猜想： $Q(4, q)$ 的 m -ovoid 仅在 $m \mid (q + 1)$ 存在 (m -ovoid 存在判据)。

The wording is identical in both languages, and the relation is written with the divisibility symbol “ $\mid$ ”. We therefore refute the literal reading:

Definition 1 (m -ovoid). Let q be a prime power and let $Q(4, q)$ be the parabolic quadric in $PG(4, q)$, regarded as a generalized quadrangle whose *generating lines* are its lines. An m -ovoid is a set of points of $Q(4, q)$ that meets every generating line in exactly m points. A 1-ovoid is simply called an *ovoid*.

The claim to be refuted is thus the necessary condition

$$\text{“an } m\text{-ovoid of } Q(4, q) \text{ exists”} \implies m \mid (q + 1). \quad (*)$$

2 The quadric $Q(4, 2)$

For $q = 2$ we use the standard vector model of $Q(4, 2)$, which is the quadric in $\text{PG}(4, 2)$ with quadratic form and polar form

$$Q(x) = x_0 + x_1x_2 + x_3x_4, \quad B(a, b) = a_1b_2 + a_2b_1 + a_3b_4 + a_4b_3$$

over $\mathbb{F}_2$ (here $x_i \in \mathbb{F}_2$ and $x_0^2 = x_0$). Thus

$$Q(a + b) = Q(a) + Q(b) + B(a, b).$$

Definition 2 (points and lines of $Q(4, 2)$). The *points* of $Q(4, 2)$ are the nonzero vectors $x \in \mathbb{F}_2^5$ with $Q(x) = 0$. Its *generating lines* are the triples

$$\{a, b, a + b\}$$

where a, b are distinct points with $B(a, b) = 0$ (equivalently, $a + b$ is again a point).

A direct enumeration gives the following well-known parameters.

Proposition 3. *$Q(4, 2)$ has exactly 15 points and 15 lines; every line contains exactly $q + 1 = 3$ points, and every point lies on exactly $q + 1 = 3$ lines.*

证明. Enumeration over the $2^5 = 32$ vectors of $\mathbb{F}_2^5$. The number of nonzero Q -isotropic vectors is $2^4 - 1 = 15$; each of the 15 points is joined to the other points b with $B(a, b) = 0$, giving 3 lines per point and $15 \cdot 3/3 = 15$ lines. See `reproduce.py` for the exact computation and `lean4/Main.lean` for the formal verification. $\square$

3 An ovoid and its complement

An ovoid of $Q(4, 2)$ has $q^2 + 1 = 5$ points. The following is one.

Theorem 4. *In the model of Definition 2, the set*

$$\mathcal{O} = \{ (0, 1, 0, 0, 0), (0, 0, 1, 0, 0), (1, 1, 1, 1, 0), (1, 1, 1, 0, 1), (0, 1, 1, 1, 1) \}$$

is an ovoid of $Q(4, 2)$: it has 5 points and meets each of the 15 generating lines in exactly 1 point. Moreover:

1. *every line of $Q(4, 2)$ meets the complement $\mathcal{O}^c$ in exactly $3 - 1 = 2$ points, so $\mathcal{O}^c$ is a 2-ovoid;*
2. $|\mathcal{O}^c| = 15 - 5 = 10 = q^2 + q$;
3. $m = 2$ *does not divide* $q + 1 = 3$.

Consequently the necessary condition $(*)$ is false, and with it conjecture 00000001006.

証明. This is a finite verification. The five listed vectors are nonzero with $Q = 0$; enumerating the 15 lines of Proposition 3 gives exactly one line through each pair of them (and no line containing two of them), so the hit count is 1 on every line. Since each line has exactly 3 points and $\mathcal{O}$ occupies exactly one of them, the complement occupies exactly the remaining 2; the complement therefore consists of $15 - 5 = 10$ points and meets every line in exactly 2 points, i.e. it is a 2-ovoid. Finally $3 = 2 \cdot 1 + 1$, so $2 \nmid 3$. All counts are tabulated in Section 4 and checked exactly by `reproduce.py` and by `lean4/Main.lean`. $\square$

Corollary 5. *Conjecture 00000001006 is false. The quadric $Q(4, 2)$ possesses a 2-ovoid although $2 \nmid (2 + 1)$.*

Remark 6 (the general complement construction). The argument is not special to $q = 2$. If $Q(4, q)$ has an ovoid $\mathcal{O}$, then $|\mathcal{O}| = q^2 + 1$ and the complement $\mathcal{O}^c$ has $(q + 1)(q^2 + 1) - (q^2 + 1) = q(q^2 + 1)$ points; every line has $q + 1$ points and meets $\mathcal{O}$ once, hence meets $\mathcal{O}^c$ exactly q times. So $\mathcal{O}^c$ is a q -ovoid, while $q \nmid q + 1$ for every $q \geq 2$. Ovoids of $Q(4, q)$ are known to exist for every prime power q (for $q = 2$ the ovoid $\mathcal{O}$ above is explicit), so the criterion $(*)$ fails for *every* $q \geq 2$ by taking $m = q$. This submission formalises the smallest instance $q = 2$; the general statement is recorded here as context.

4 All 15 lines and their hit counts

Each line of $Q(4, 2)$ is listed as $\{a, b, a + b\}$ in coordinates $(x_0, x_1, x_2, x_3, x_4)$; the last two columns give the number of points of the line lying in $\mathcal{O}$ and in $\mathcal{O}^c$.

generating line	$ \ell \cap \mathcal{O} $	$ \ell \cap \mathcal{O}^c $
$\{(0, 1, 0, 0, 0), (0, 0, 0, 1, 0), (0, 1, 0, 1, 0)\}$	1	2
$\{(0, 1, 0, 0, 0), (0, 0, 0, 0, 1), (0, 1, 0, 0, 1)\}$	1	2
$\{(0, 1, 0, 0, 0), (1, 0, 0, 1, 1), (1, 1, 0, 1, 1)\}$	1	2
$\{(0, 0, 1, 0, 0), (0, 0, 0, 1, 0), (0, 0, 1, 1, 0)\}$	1	2
$\{(0, 0, 1, 0, 0), (0, 0, 0, 0, 1), (0, 0, 1, 0, 1)\}$	1	2
$\{(0, 0, 1, 0, 0), (1, 0, 0, 1, 1), (1, 0, 1, 1, 1)\}$	1	2
$\{(1, 1, 1, 0, 0), (0, 0, 0, 1, 0), (1, 1, 1, 1, 0)\}$	1	2
$\{(1, 1, 1, 0, 0), (0, 0, 0, 0, 1), (1, 1, 1, 0, 1)\}$	1	2
$\{(1, 1, 1, 0, 0), (1, 0, 0, 1, 1), (0, 1, 1, 1, 1)\}$	1	2
$\{(0, 1, 0, 1, 0), (0, 0, 1, 0, 1), (0, 1, 1, 1, 1)\}$	1	2
$\{(0, 1, 0, 1, 0), (1, 1, 1, 0, 1), (1, 0, 1, 1, 1)\}$	1	2
$\{(0, 0, 1, 1, 0), (0, 1, 0, 0, 1), (0, 1, 1, 1, 1)\}$	1	2
$\{(0, 0, 1, 1, 0), (1, 1, 1, 0, 1), (1, 1, 0, 1, 1)\}$	1	2
$\{(1, 1, 1, 1, 0), (0, 1, 0, 0, 1), (1, 0, 1, 1, 1)\}$	1	2
$\{(1, 1, 1, 1, 0), (0, 0, 1, 0, 1), (1, 1, 0, 1, 1)\}$	1	2
total over all lines	15	30

Every row has hit counts 1 and 2, so $\mathcal{O}$ is an ovoid and $\mathcal{O}^c$ is a 2-ovoid. (The 15 lines above are exactly the whole line set; this completeness is verified by enumeration in `reproduce.py` and by `lines15_is_all_lines` in `lean4/Main.lean`.)

5 The divisibility failure

The only arithmetic ingredient is

$$q + 1 = 3, \quad m = 2, \quad 3 = 2 \cdot 1 + 1 \implies 2 \nmid 3.$$

So an m -ovoid exists for $m = 2$, but m fails to divide $q + 1$. This contradicts $(*)$ directly.

6 Caveats and the exact reading

- **The literal reading is the divisibility one.** Both the English and the Chinese statements write “ $m \mid (q + 1)$ ”, i.e. “ m divides $q + 1$ ”. The counterexample above refutes precisely this reading.
- **Weaker readings.** If a reader were to replace “ $m \mid (q + 1)$ ” by “ $m \leq q + 1$ ”, then $m = 2 \leq 3$ would be compatible with the claim and the witness would not apply. But that is not what the filed statement says; the file contains no inequality and no other condition. We record this only to make the reading explicit.
- **Only “only when” is used.** The statement is one-directional (a necessary condition),

which is exactly what (*) formalises; the counterexample violates that necessary condition, so it refutes the conjecture under either the one-directional or the biconditional reading.

- **No algorithmic subtlety.** The ovoid $\mathcal{O}$ and its complement are given by explicit vectors; every claim is a finite, exact computation with no search heuristics or probabilistic steps.
- **Relation to the literature.** $Q(4, q)$ is a classical generalized quadrangle of order (q, q) ; its ovoids (respectively, the m -ovoids it is known to admit) are a standard topic. Nothing in this note depends on any unpublished result: the $q = 2$ instance is fully explicit.

7 Formalisation and reproducibility

The Lean 4 project under `lean4/` (core Lean only, `import Std`, no `Mathlib`, no `sorry`) models points as `Fin 5 → Bool`, defines Q , B and the line relation, hardcodes the ovoid $\mathcal{O}$, and closes by `decide`:

```
theorem O_is_ovoid : IsMOvoid 1 0 = true
theorem Ocomp_is_2ovoid : IsMOvoid 2 Ocomp = true
theorem two_not_dvd_three : ¬ (2 | 3)
theorem two_ovoid_exists : ∃ S, IsMOvoid 2 S = true
theorem lines15_is_all_lines : allLinesE = lines15
```

The axiom audit `lake env lean Check.lean` reports no `sorryAx` (the finite facts depend at most on `propext`). The script `reproduce.py` (Python 3 standard library only) rebuilds $Q(4, 2)$, enumerates all six ovoids, verifies the hit counts of $\mathcal{O}$ and $\mathcal{O}^c$, and exits non-zero on any failure.

```
$ python3 reproduce.py
... PASS: Q(4,2) has a 2-ovoid ... conjecture 00000001006 is FALSE.
$ cd lean4 && lake build && lake env lean Check.lean
$ tectonic main.tex --outdir build
```